Queueing analysis of a battery-backed electrical vehicle charging system

ABSTRACT

We study a battery-buffered electric vehicle fast charging station modeled as a piecewise-deterministic Markov process. A battery mediates between the grid connection and one or more chargers, enabling fast service when the battery is charged and falling back to the grid rate when depleted. For the single–server loss system we derive the stationary distribution in closed form, uncovering a decoupling property whereby the charger state and battery level are independent in steady state, and obtain explicit formulas for several performance metrics. We extend the analysis to the multi–server loss system, showing that the stationary densities retain an exponential form with occupancy weights governed by a simple recurrence, related to the Erlang-B formula.

Keywords

Piecewise deterministic Markov processes, Electrical Vehicles.

1. INTRODUCTION

The rapid adoption of electric vehicles (EVs) is placing increasing demands on electrical infrastructure. A key challenge is the provisioning of public *fast charging*: modern DC fast chargers can deliver 150 kW or more per port, requiring grid connections sized for their peak output. Yet in practice a charging station is not used continuously; vehicles arrive stochastically, and the chargers are idle most of the time. Sizing the grid connection for worst-case simultaneous demand is therefore wasteful, civil infrastructure becomes costly, and utility demand charges (fees based on peak power draw) can represent the dominant operating cost of a fast charging site [8]. A battery, acting as an energy buffer, placed between the grid and the chargers resolves this mismatch: the battery gradually charges from the grid at a steady, lower rate and discharges at high power during the brief intervals a vehicle is present, enabling a much smaller and cheaper grid connection. While battery storage at EV stations has been studied before, it has been done primarily in the context of renewable energy integration, where the battery smooths intermittent solar or wind supply [1,6,10]. However, with the advent of fast DC chargers available in highways and remote areas, placing a battery buffer in these remote locations with shallow grid access has become a compelling idea.

IFIP Performance '26. 3-5 November 2026, Ghent, Belgium

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The central analytical question this raises is a queueing one: given that vehicles arrive randomly, and the battery recharges during idle or low utilization periods, how often does a vehicle arrive to find the battery depleted, forcing service at the slower grid rate? This coupling between the continuous battery state and the discrete service rate is what distinguishes the battery-buffered station from a standard queue, and it has not been captured in the existing EV charging literature, which models charging with a fixed service rate [2, 5, 9].

In this paper, we propose a station model using a Piecewise-Deterministic Markov Process (PDMP) [3, 4]: the battery level evolves deterministically between random events, while car arrivals and departures trigger instantaneous switches. The state is then the pair (n, x) of system occupancy and battery level. We derive the generator of this process, and compute the stationary distribution in closed form for the single-server loss system. A key structural finding is a *decoupling property*: in steady state the charger occupancy and battery level are conditionally independent given the battery is the interior of the state space (partially charged), so that the stationary density factors as a product of an exponential in x and a mode-dependent weight. This yields explicit formulas for the blocking probability, mean charging time, and a *slowdown factor* quantifying the service degradation as a function of a car's energy demand. We then extend the analysis to a loss system with s parallel chargers sharing a common battery. The exponential form of the stationary density is preserved, with the mode weights now governed by a simple recurrence. Armed with these formulas, we discuss several performance issues, design objectives that provide guidelines on how to design such systems and provide some numerical examples.

2. SINGLE-SERVER MODEL

We consider a single EV charging station consisting of one charger connected to the electrical grid, equipped with a battery buffer of capacity $B > 0$. Cars arrive according to a Poisson process of rate $\lambda > 0$, each carrying an energy demand $Y \sim \mathrm{Exp}(\mu)$. The station operates as a *loss system*: a car that arrives when the charger is busy is turned away.

The system parameters are the following: the grid may supply a maximum power p_1, and this energy may be used to charge the battery or the car, if present. When the system is idle, the battery is recharged at rate $0 < p_0 \leqslant p_1$. Inequality reflects that we may want to recharge the battery at a slower rate to preserve battery life. The EV charging station itself may provide power $p_2 > p_1$, which amounts to fast charging.

When a car is present, this power is provided by the grid at rate p_1 and the extra amount $c := p_2 - p_1 > 0$ is provided by the battery, whenever is not empty. If the battery empties, the system reverts to charging only from the grid at rate p_1, and the battery stays depleted until the current car finishes service. We denote by $x(t) \in [0, B]$ the current state of the battery at time t, and by $n(t) \in \{0, 1\}$ the presence of not of a charging vehicle.

Under the above assumptions, when a car is admitted, service proceeds in at most two phases. In *Phase 1* (fast charging), while the battery level $x > 0$, the car charges at rate p_2 and the battery discharges at rate c. In *Phase 2* (slow charging), once the battery is exhausted, $x = 0$ and the car charges at rate p_1, while x remains at zero. A car with small demand may complete service entirely in Phase 1 if its service requirement is low enough. Under the exponential energy demands assumed before, the instantaneous departure rate from the system is therefore $\mu p(x)$, where:

$$p(x) = p_2 \, \mathbf{1}_{x>0} + p_1 \, \mathbf{1}_{x=0}. \tag{1}$$

The system state at time t is thus the pair $Z(t) = (n(t), x(t))$. Between arrival and departure events, the battery evolves deterministically according to $\dot{x} = r(n, x)$, where

$$r(n,x) = \begin{cases} p_0 & n = 0,\ x < B, \\ 0 & n = 0,\ x = B, \\ -c & n = 1,\ x > 0, \\ 0 & n = 1,\ x = 0. \end{cases} \tag{2}$$

The jumps are spontaneous events: *arrivals* at $(0, x)$ occur at rate λ and the states transitions (deterministically) to $(1, x)$; *departures* at $(1, x)$ with $x > 0$ occur at rate μp_2 and transition to $(0, x)$; *departures* at $(1, 0)$ occur at rate μp_1 and transition to $(0, 0)$, or in short $(1, x) \mapsto (0, x)$ at rate $\mu p(x)$. In all cases the battery level is unchanged at the jump epoch.

The process $Z(t)$ with state space $\{0, 1\} \times [0, B]$ fits the PDMP framework of Davis [3,4], though the standard theory requires an open state space and forces the process to jump before reaching the boundary. Neither condition holds here: E is compact, and the flow reaches $x = 0$ and $x = B$ in finite time without a jump. Nevertheless, the vector field (2) is bounded and monotone on each mode, so the flow is globally well-defined. The strong Markov property follows from the martingale-problem approach, which extends to this setting since all jump rates are bounded and the jump kernel leaves x unchanged, i.e. jumps in the continuous variable are bounded.

The extended generator $\mathcal{A}$ of $Z(t)$ has domain $\mathcal{D}(\mathcal{A})$ consisting of functions $\psi : E \to \mathbb{R}$ such that $\psi(n, \cdot)$ is absolutely continuous on $[0, B]$ for each n. For $\psi \in \mathcal{D}(\mathcal{A})$,

$$\mathcal{A}\psi(0,x) = r(0,x)\,\psi_0'(x) + \lambda\,[\psi(1,x) - \psi(0,x)], \tag{3a}$$

$$\mathcal{A}\psi(1,x) = r(1,x)\,\psi_1'(x) + \mu p(x)\,[\psi(0,x) - \psi(1,x)], \tag{3b}$$

where $\psi_n' = \partial_x \psi(n, \cdot)$ is defined Lebesgue-a.e. The first term in each line is the flow contribution; the second is the jump contribution, which in fact just switches mass from $n = 0$ to $n = 1$ and vice versa leaving x unchanged. For any $\psi \in \mathcal{D}(\mathcal{A})$, the process $M_t^\psi := \psi(Z(t)) - \psi(Z(0)) - \int_0^t \mathcal{A}\psi(Z(s))\,ds$ is a local martingale [3], and the process is right-continuous and a strong Markov process.

Note that an arriving car always finds $x > 0$ with probability one. During any idle period the battery charges at rate $p_0 > 0$, and since the idle time before the next arrival is Exp(λ) (positive a.s.), the battery is strictly positive at every arrival epoch. The state $(n = 0, x = 0)$ therefore should have zero probability under any stationary distribution.

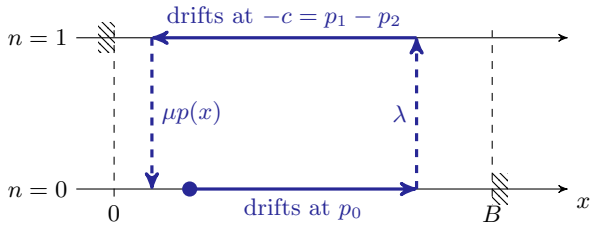


Figure 1: State space and system dynamics, showing a typical trajectory of $Z(t)$.

The adjoint operator $\mathcal{A}^*$ operates on the probability distributions on E and provides the forward Kolmogorov equation of the system [4]. A stationary distribution π satisfies $\int \mathcal{A}\psi \, d\pi = 0$ for all $\psi \in \mathcal{D}(\mathcal{A})$, which is the weak form of the Kolmogorov forward equation $\mathcal{A}^*\pi = 0$. We now proceed to solve the steady state of this system by positing a particular form of π using the characteristics of the system.

3. SINGLE-SERVER IN STEADY-STATE

3.1 Stationary distribution

Since E is compact and $Z(t)$ is irreducible (every open rectangle $\{n\} \times (a, b)$ is reachable with positive probability from any starting point, by a combination of deterministic drift and Poisson events), the process is ergodic with a unique stationary distribution π [7].

We now seek π as a mixed measure with densities on $(0, B)$ and point masses at the boundary points where flow accumulates. Therefore, we propose the following solution structure:

$$\pi = f_0(x)\,dx + f_1(x)\,dx + \pi_{0,B}\,\delta_{(0,B)} + \pi_{1,0}\,\delta_{(1,0)}. \tag{4}$$

Here $f_n(x)$ $n = 0, 1$ are the interior densities and $\pi_{0,B}, \pi_{1,0} \geq 0$ are point masses at the corners where the flow accumulates: $x = B$ in mode 0 (since $r(0, \cdot) = p_0 > 0$) and $x = 0$ in mode 1 (since $r(1, \cdot) = -c < 0$). The other two corners should carry no atoms since flow arriving on them is immediately carried into the interior by the flow (cf. Figure 1).

Substituting this form into $\int \mathcal{A}\psi \, d\pi = 0$ and using eqs. (3), the stationarity condition $\int \mathcal{A}\psi \, d\pi = 0$ for all $\psi \in \mathcal{D}(\mathcal{A})$ yields the following balance equations for the interior densities:

$$p_0\,f_0'(x) = -\lambda\,f_0(x) + \mu p_2\,f_1(x), \tag{5a}$$

$$-c\,f_1'(x) = \quad \lambda\,f_0(x) - \mu p_2\,f_1(x). \tag{5b}$$

As for the atoms, we obtain the following *boundary flux equations*, obtained by flux conservation at each boundary point. At $x = 0$ in mode 1, the inward flow $c\,f_1(0^+)$ feeds the atom $\pi_{1,0}$, which drains via departures at rate μp_1:

$$c\,f_1(0^+) = \mu p_1\,\pi_{1,0}. \tag{5c}$$

Similarly, at $x = B$ in mode 0, the inward flow $p_0\,f_0(B^-)$ feeds the atom $\pi_{0,B}$, which drains via arrivals at rate λ:

$$p_0\,f_0(B^-) = \lambda\,\pi_{0,B}. \tag{5d}$$

Finally, at $(n, x) = (0, 0)$, mass arriving from departures at rate μp_1 must immediately be carried at rate p_0 into the interior (no atom), yielding:

$$p_0\, f_0(0^+) = \mu p_1\, \pi_{1,0}. \tag{5e}$$

And finally by a similar argument, at $(n, x) = (1, B)$

$$c\, f_1(B^-) = \lambda\, \pi_{0,B}. \tag{5f}$$

Finally we have the normalization condition $\int_0^B (f_0 + f_1)\, dx + \pi_{0,B} + \pi_{1,0} = 1$. We can now state the main result of this section.

THEOREM 3.1. *Let $\theta := \mu p_2/c - \lambda/p_0$. The unique stationary distribution of $Z(t)$ has the form (4) with*

$$f_0(x) = \frac{e^{\theta x}}{K}, \qquad f_1(x) = \frac{p_0}{c} \cdot \frac{e^{\theta x}}{K}, \qquad x \in (0, B), \tag{6a}$$

$$\pi_{1,0} = \frac{p_0}{\mu p_1\, K}, \qquad \pi_{0,B} = \frac{p_0\, e^{\theta B}}{\lambda\, K}, \tag{6b}$$

and the normalization constant is

$$K = \frac{p_0 + c}{c} \cdot \frac{e^{\theta B} - 1}{\theta} + \frac{p_0}{\mu p_1} + \frac{p_0\, e^{\theta B}}{\lambda}, \tag{7}$$

with the convention $(e^{\theta B} - 1)/\theta = B$ when $\theta = 0$.

PROOF SKETCH. Adding (5a), (5b) gives $\frac{d}{dx}[p_0 f_0 - c f_1] = 0$, so $p_0 f_0 - c f_1$ is constant on $(0, B)$. Equations (5c) and (5e) together give $p_0 f_0(0^+) = c f_1(0^+)$, so the constant is zero. Therefore, the densities should be proportional to each other, i.e. $f_1(x) = (p_0/c)\, f_0(x)$ for all $x \in (0, B)$. Substituting this condition into (5a) yields a first order differential equation for f_0, which accounting for the definition of θ is simply $f_0'(x) = \theta\, f_0(x)$. We conclude that f_0 (and thus also f_1) should be exponential: $f_0(x) = f_0(0^+)\, e^{\theta x}$. With this expression, the atoms follow directly from (5d), (5e), and (5f) is automatically satisfied, confirming consistency. Setting $f_0(0^+) = 1/K$ and imposing normalization determines K as in (7). □

The parameter θ has a physical interpretation: define the offered load of the system $\rho := \lambda/\mu$, i.e. the average external power request. Consider now a single idle-busy cycle in the absence of battery boundaries: the battery charges at rate p_0 during the idle period (mean duration $1/\lambda$) and discharges at rate c during the busy period with mean duration $1/(\mu p_2)$. Then, the expected net battery change per cycle is therefore $\Delta := \frac{p_0}{\lambda} - \frac{c}{\mu p_2}$. It is easy to see that θ has the same sign as Δ, in particular:

$$\theta > 0 \Leftrightarrow \rho < \frac{p_0 p_2}{c} = \rho \frac{p_0 p_2}{p_2 - p_1}$$

This corresponds to a *lightly loaded* station, where the battery gains energy on average on each cycle and thus tends to fill, concentrating its mass towards $x = B$. When $\lambda/\mu > p_0 p_2/c$, we are in the *highly loaded* regime, $\theta < 0$, the battery tends to drain and mass concentrates near $x = 0$, where a significant fraction of service occurs at the slow rate p_1. When $\theta = 0$ the battery neither drains nor fills on average and stationary density is uniform on $(0, B)$.

3.2 Performance metrics

We now derive the key performance metrics directly from Theorem 3.1. We begin by computing the blocking probability, that due to the PASTA property, is just the stationary probability of being in state 1, i.e.

$$P_B = \int_0^B f_1(x)\, dx + \pi_{1,0} = \frac{p_0}{K} \left(\frac{e^{\theta B} - 1}{c\,\theta} + \frac{1}{\mu p_1} \right). \tag{8}$$

Note that this probability is also the probability that the charger is working. As $B \to 0$, $P_B \to \rho/(\rho + p_1)$, the blocking probability of an $M/M/1/1$ queue at rate p_1, since cars can only be served from the grid. Conversely, if the battery is large, as $B \to \infty$ we distinguish two cases: in the lightly loaded regime, $\theta > 0$ and $P_B \to \rho/(\rho + p_2)$, corresponding to service always at the fast rate p_2 because the battery can cope with the load and the system operates most of the time in fast charging. However, if the system is highly loaded ($\theta < 0$), P_B converges to a value strictly between these two limits even as $B \to \infty$, reflecting that the system operates *partly in both charging regimes* because the battery regularly depletes.

We now focus on the mean service time. By Little's law, since there is only one server working and arrivals onto a busy system are blocked, we have $E[N] = P_B = \lambda(1 - P_B)E[T]$ and therefore, the mean service time of an admitted car is:

$$\mathbb{E}[T] = \frac{P_B}{\lambda(1 - P_B)}. \tag{9}$$

A more meaningful quantity is the *average slowdown*, which we define as $E[T]$ normalized by the average service time in a system that operates at the fast rate p_2 permanently, which is given by:

$$\bar{S} := \frac{E[T]}{E[Y]} = \frac{\mu p_2 P_B}{\lambda(1 - P_B)} = \frac{p_2}{\rho} \frac{P_B}{1 - P_B} \tag{10}$$

A final quantity of interest is how different demands are affected in terms of charging time. One expects that larger demands are more affected since the battery may deplete during service. To this end, define the *slowdown factor*:

$$S(y) := \frac{\mathbb{E}[T \mid Y = y]}{(y/p_2)}. \tag{11}$$

The following proposition is proved in the Appendix.

PROPOSITION 3.2. *Let $u = cy/(p_2 B) = y/x^*$ be the demand normalised by the critical threshold $x^* = p_2 B/c$, so that $u < 1$ means the battery may survive the charge and $u \geq 1$ means it always depletes. Then:*

$$S(u) = 1 + \frac{1}{p_1(B + p_0/\lambda)} \left[\frac{e^{\theta u B} - 1}{\theta u B} - 1 \right], \qquad u < 1, \tag{12a}$$

$$S(u) = 1 + \frac{1}{p_1(B + p_0/\lambda)} \Big\{ \Big(1 - \tfrac{1}{u}\Big) \Big[\tfrac{e^{\theta B}}{\theta} + \tfrac{p_0}{\lambda} e^{\theta B} \Big] + \tfrac{e^{\theta B} - 1}{\theta^2 u} - \tfrac{1}{\theta} \Big\}, \qquad u \geq 1. \tag{12b}$$

In both cases $S(u) \geq 1$, with $S \to 1$ as $u \to 0$ or $B \to \infty$, and $S \to p_2/p_1$ as $u \to \infty$ or $B \to 0$.

3.3 Optimal battery regime

For practical battery health it is desirable to charge the battery *slowly*, and to keep it *cycling* continuously rather

maintaining it fully charged or fully depleted for long periods. The stationary distribution of Theorem 3.1 reveals a natural design criterion that achieves both goals simultaneously: namely control the value of the idle recharge time to reduce recharging power, but aim for the balanced regime where $\theta = 0$ so the battery is kept most of the time in $(0, B)$ and not the boundaries.

To analyze this, we first introduce the following definitions. Let $\alpha = p_2/p_1 > 1$ be the *speed-up ratio* of fast vs. slow charging. Let $\beta = \mu B p_2/c$ be the *normalized battery capacity*, the ratio between the battery drain time and the average fast recharge session duration, i.e. how many full fast-charge sessions the battery can sustain before depleting. Finally, define the *normalized load* $\tilde{\rho} = \rho/p_2 = \lambda/(\mu p_2)$.

We aim for the uniform stationary density achieved when $\theta = 0$. Setting $\theta = \mu p_2/c - \lambda/p_0 = 0$ uniquely determines the idle recharge rate as:

$$p_0^* = \frac{\lambda c}{\mu p_2} = \tilde{\rho}\, c. \tag{13}$$

The constraint $p_0^* \leqslant p_1$ requires then:

$$\tilde{\rho} \leqslant \frac{p_1}{c} = \frac{1}{\alpha - 1} \iff \rho \leqslant \rho_{\max} = \frac{p_1 p_2}{c}. \tag{14}$$

Below this threshold the operator can minimize battery stress by throttling down the idle recharge rate (slower recharge) and achieve the uniform regime. Above it, $\theta < 0$ regardless of p_0 and the battery drains on steady state.

Setting now $\theta = 0$ and $p_0 = p_0^*$ in the expressions for K and P_B, one finds a simple form for the average slowdown in terms of the normalized battery capacity:

$$\bar{S} = \frac{\alpha + \beta}{\beta + 1}. \tag{15}$$

Remarkably, this depends only on α and β and not on the load. This is very useful from a design perspective since *the same design* works across all (feasible) loads $\rho < \rho^*$, and decreases from $\alpha = p_2/p_1$ at $\beta \to 0s$ (no battery) to 1 at $\beta \to \infty$. Setting a target slowdown $\bar{S}_0 \in (1, \alpha)$ we get:

$$\beta^* = \frac{\alpha - \bar{S}_0}{\bar{S}_0 - 1} \iff B^* = \frac{c}{\mu p_2} \frac{\alpha - \bar{S}_0}{\bar{S}_0 - 1}. \tag{16}$$

Finally the blocking probability becomes:

$$P_B = \frac{\tilde{\rho}\bar{S}_0}{1 + \tilde{\rho}\bar{S}_0} = \frac{\tilde{\rho}(\alpha + \beta)}{\beta + 1 + \tilde{\rho}(\alpha + \beta)}. \tag{17}$$

This has the form of an $M/M/1/1$ blocking formula with effective load $\tilde{\rho}\bar{S}$, since $P_B = \tilde{\rho}\bar{S}/(1 + \tilde{\rho}\bar{S})$. Unlike $\bar{S}$, the blocking probability depends on $\tilde{\rho}$, as expected.

4. THE MANY-SERVER CASE

We now extend the model to $s \geq 1$ parallel chargers sharing a single battery buffer of capacity B and a grid connection of rate p_1, still operating as a loss system. We impose $p_1 < sp_2$ (the grid alone cannot sustain full occupancy) and, for simplicity, set $p_0 = p_1$ (the idle recharge rate equals the grid supply rate). This last condition ensures that the net battery drift $r_n = p_1 - np_2$ has a uniform expression for all $n \geq 0$: the battery charges at rate $r_n > 0$ whenever $n \leq n^* := \lfloor p_1/p_2 \rfloor$ (including idle), and drains at rate $|r_n| = np_2 - p_1 > 0$ when $n > n^*$.[1] Relaxing $p_0 \neq p_1$ is possible but breaks the uniform structure of r_n and complicates the recurrence below without changing its qualitative form.

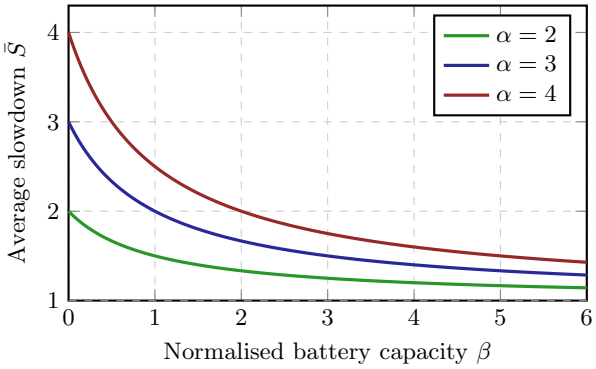


Figure 2: Average slowdown $\bar{S}$ in the optimal battery regime as a function of normalized capacity $\beta = \mu B p_2/c$, for three values of the speed-up ratio $\alpha = p_2/p_1$.

The state space is $E = \{0, \ldots, s\} \times [0, B]$, compact. Arrivals occur at rate λ when $n < s$; departures occur at total rate $n\mu p_2$ when $x > 0$ and at rate μp_1 when $x = 0$ (the grid supplies p_1 shared equally among the n busy chargers, giving total throughput μp_1 regardless of n). The process $Z(t) = (n(t), x(t))$ is again an ergodic PDMP by the same arguments as Section 2. The stationary distribution has densities $f_n(x)$ on $(0, B)$ and point masses at (n, B) for $n \leq n^*$ and $(n, 0)$ for $n > n^*$.

THEOREM 4.1. *Let $\rho = \lambda/(\mu p_2)$. There exists a unique $\theta \in \mathbb{R}$, satisfying the boundary flux equations at $x = 0$ and $x = B$ together with normalization, such that the stationary densities take the form*

$$f_n(x) = C\, a_n(\theta)\, e^{\theta x}, \qquad x \in (0, B),\ n = 0, \ldots, s, \tag{18}$$

where $C > 0$ is a normalization constant and the coefficients $a_n(\theta)$ satisfy $a_0 = 1$ and

$$a_{n+1} = \frac{\theta\, S_n + \lambda\, a_n}{(n+1)\mu p_2}, \quad S_n = \textstyle\sum_{k=0}^{n} r_k\, a_k, \quad n = 0, \ldots, s-1. \tag{19}$$

The proof is given in Appendix B.

Performance metrics follow from C and the point masses obtained in the proof via the same formulas as Section 3.2.

5. SIMULATION

6. CONCLUSIONS

We have studied a battery-buffered EV fast charging station using the framework of piecewise-deterministic Markov processes. For the single-server loss system we derived the stationary distribution in closed form, establishing a decoupling property that separates the battery level from the charger state in steady state, and obtained explicit expressions for

[1] If $p_2/p_1 \in \mathbb{Z}$ then $r_{n^*} = 0$ exactly: at occupancy n^* the battery neither charges nor drains. The balance equation (22) at $n = n^*$ then contains no derivative term and f_{n^*} is determined algebraically; the exponential form and recurrence (19) remain valid without modification.

the blocking probability, mean charging time, and slowdown factor. We extended the analysis to a loss system with s parallel chargers sharing a common battery, showing that the stationary densities retain an exponential form in the battery level with occupancy-dependent weights governed by a simple recurrence. In both cases the key parameter θ, the sign of the mean battery drift, determines whether the system operates in a battery-rich or battery-scarce regime and controls the deviation of the blocking probability from the classical Erlang-B value.

The analysis highlights a structural difference from standard fluid queues: the state-dependent service rate at $x = 0$ couples the discrete and continuous components in a way that prevents a product-form solution in the many-server case while still admitting an exponential form in x. Numerical illustrations of the performance metrics and a study of the optimal battery sizing problem are left for future work.

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APPENDIX

A. PROOF OF PROPOSITION 3.2

Given admission at battery level $x \in (0, B)$ with demand $Y = y$, service runs at rate p_2 until the battery depletes at time x/c, then at rate p_1 for any remaining demand. The total charging time is

$$T(x, y) = \frac{y}{p_2} + \left(\frac{1}{p_1} - \frac{1}{p_2}\right) \max\Big(y - \frac{p_2 x}{c},\, 0\Big). \qquad (20)$$

By PASTA, an admitted car sees the stationary distribution conditioned on $n = 0$, so the admitted battery level has density $f_0(x)/(1 - P_B)$ on $(0, B)$ plus atom $\pi_{0,B}/(1 - P_B)$ at $x = B$. Writing $x_y = cy/p_2$ for the battery level that would be exactly depleted by demand y, substituting (20) and using $p_2 - p_1 = c$ gives

$$\begin{aligned}\mathbb{E}[T \mid Y = y] &= \frac{y}{p_2} \\ &\quad + \frac{c}{p_1 p_2 (1 - P_B)} \Big[\int_0^{\min(x_y, B)} \Big(y - \tfrac{p_2 x}{c}\Big) f_0(x)\, dx \\ &\qquad + \mathbf{1}_{y \ge p_2 B/c}\, \pi_{0,B} \left(y - \tfrac{p_2 B}{c}\right) \Big]. \qquad (21)\end{aligned}$$

We now substitute $u = y/x^* = cy/(p_2 B)$, so $y = up_2 B/c$ and $x_y = uB$, and use $f_0(x) = e^{\theta x}/K$ and $K(1 - P_B) = B + p_0/\lambda$.

For $u < 1$, the atom at $x = B$ contributes nothing and the integral runs to uB. Denoting $\mathrm{I} = \int_0^{uB} (y - p_2 x/c)\, e^{\theta x}\, dx/K$, integration by parts gives

$$\begin{aligned}\mathrm{I} &= \frac{p_2}{cK} \left[uB \cdot \frac{e^{\theta uB} - 1}{\theta} - \frac{uB\, e^{\theta uB}}{\theta} + \frac{e^{\theta uB} - 1}{\theta^2} \right] \\ &= \frac{p_2 B}{cK} \cdot \frac{e^{\theta uB} - 1 - \theta uB}{\theta^2}.\end{aligned}$$

Substituting into (21), dividing by $y/p_2 = uB/c$ and simplifying using $K(1 - P_B) = B + p_0/\lambda$ yields (12a).

For $u \ge 1$, the integral runs to B and the atom contributes. Writing $\pi_{0,B} = p_0 e^{\theta B}/(\lambda K)$ and denoting

$$\begin{aligned}\mathrm{II} &= \int_0^B \Big(y - \frac{p_2 x}{c}\Big) \frac{e^{\theta x}}{K}\, dx \\ &= \frac{p_2}{cK} \left[uB \cdot \frac{e^{\theta B} - 1}{\theta} - \frac{B e^{\theta B}}{\theta} + \frac{e^{\theta B} - 1}{\theta^2} \right],\end{aligned}$$

the atom term equals $(p_0/\lambda)\, e^{\theta B}(u - 1)B/(B + p_0/\lambda)$. Collecting, dividing by $y/p_2 = uB/c$, and using $K(1 - P_B) = B + p_0/\lambda$ yields (12b).

The limits $S \to 1$ as $u \to 0$ or $B \to \infty$ and $S \to p_2/p_1$ as $u \to \infty$ or $B \to 0$ follow directly from the formulas. □

B. PROOF OF THEOREM 4.1

The stationarity condition $\int \mathcal{A}\psi\, d\pi = 0$ applied to the interior densities yields, for $0 \le n \le s$,

$$\begin{aligned}r_n f_n'(x) &= \lambda f_{n-1}(x)\, \mathbf{1}_{n>0} + (n+1)\mu p_2\, f_{n+1}(x)\, \mathbf{1}_{n<s} \\ &\quad - (\lambda\, \mathbf{1}_{n<s} + n\mu p_2)\, f_n(x), \quad x \in (0, B). \qquad (22)\end{aligned}$$

Summing from $k = 0$ to n for $0 \le n \le s - 1$, the right side telescopes to $-\lambda f_n + (n + 1)\mu p_2 f_{n+1}$, giving the summed

equations

$$\sum_{k=0}^{n} r_k f_k'(x) = -\lambda f_n(x) + (n+1)\mu p_2\, f_{n+1}(x). \tag{23}$$

Substituting $f_n(x) = C a_n e^{\theta x}$ into (23) and cancelling $Ce^{\theta x}$ yields the recurrence (19).

Boundary flux equations. At $x = B$, for each $n \le n^*$ (where $r_n > 0$, flow pushes toward $x = B$), flux balance gives:

$$r_n\, f_n(B^-) = -\lambda\, \pi_n^{(B)} + \mu p_2\, (n+1)\, \pi_{n+1}^{(B)}\, \mathbf{1}_{n<n^*} + \lambda\, \pi_{n-1}^{(B)}\, \mathbf{1}_{n>0}. \tag{24}$$

At $x = 0$, for each $n > n^*$ (where $r_n < 0$, flow pushes toward $x = 0$), flux balance gives:

$$|r_n|\, f_n(0^+) = -(\lambda\, \mathbf{1}_{n<s} + \mu p_1)\, \pi_n^{(0)} + \mu p_1\, \pi_{n+1}^{(0)}\, \mathbf{1}_{n<s} + \lambda\, \pi_{n-1}^{(0)}\, \mathbf{1}_{n>n^*+1}. \tag{25}$$

Additionally, at $x = 0$ for $n = n^* + 1$ the atom $\pi_{n^*+1}^{(0)}$ receives inflow from mode n^* via the interior flow reaching $x = 0$ and is fed back into the interior at mode n^* (no atom there since $r_{n^*} \ge 0$):

$$r_{n^*}\, f_{n^*}(0^+) = \mu p_1\, \pi_{n^*+1}^{(0)} - \lambda\, f_{n^*}(0^+)\, \mathbf{1}_{n^*>0}. \tag{26}$$

Determining θ and normalization. For each θ, the recurrence (19) uniquely determines $a_1, \ldots, a_s$ given $a_0 = 1$. The point masses $\pi_n^{(B)}$ and $\pi_n^{(0)}$ are then obtained from (24)–(26) by forward substitution, giving each as a continuous function of θ. Substituting into the normalization condition

$$C \sum_{n=0}^{s} a_n(\theta)\, \frac{e^{\theta B} - 1}{\theta} + \sum_{n=0}^{n^*} \pi_n^{(B)}(\theta) + \sum_{n=n^*+1}^{s} \pi_n^{(0)}(\theta) = 1 \tag{27}$$

yields a scalar equation for θ. A continuity and sign argument shows this equation has a unique root, which is the θ asserted in the theorem.

Consistency with $s = 1$. For $s = 1$ and $p_0 = p_1$, we have $n^* = 0$, $r_0 = p_1$, $r_1 = -c$. The recurrence gives $a_1 = (\theta p_1 + \lambda)/(\mu p_2)$. The boundary condition at $x = 0$ forces $c\, f_1(0^+) = \mu p_1\, \pi_1^{(0)}$, i.e. $c a_1 = \mu p_1 \cdot p_1 a_1 \cdot$ (from atom relation), which together with the boundary at $x = B$ gives $\theta = \mu p_2/c - \lambda/p_1$, recovering Theorem 3.1. $\square$